



UNIT PURPOSE

Students learn how measurements become defensible evidence. The unit moves from tools, units, and notation to factor-label reasoning, density, percent error, and evidence-based claims about accuracy and precision.

INSTRUCTIONAL RHYTHM

Engage -> Teach -> Model -> Guided Practice -> Independent Practice -> Exit Ticket -> Mixed Review -> Unit Test

SUGGESTED PACING

- 1 Lesson 1: Measurement Evidence and SI Records. Use Slides 1-3, Student Pages 1-2, and the exit ticket.
- 2 Lesson 2: Scientific Notation and Significant Figures. Model rounding only after the full calculation.
- 3 Lesson 3: Dimensional Analysis. Require units on every factor and a visible cancellation check.
- 4 Lesson 4: Density and Percent Error. Connect calculations to measured evidence and units.
- 5 Lesson 5: Accuracy, Precision, and Error. Compare repeated trials before making a claim.
- 6 Cumulative mixed review, feedback, and targeted reteaching.
- 7 Unit Test A. Reserve Form B for a retake, alternate class, or test security.

PACKAGE MAP

Projectable slides teach the lesson. Student pages provide guided notes, modeled thinking, supported practice, independent practice, and exit tickets. Tests A and B follow all instruction and review. The teacher file contains every answer and rationale.



Support, monitor, and respond

BEFORE TEACHING

Decide calculator expectations and the significant-figure convention used by your course. Confirm students can multiply powers of ten. Preview the pictured tools and remind students that the correct tool depends on the quantity and needed resolution.

FORMATIVE LOOK-FORS

Students attach units to measured values, distinguish exact conversion factors from measured data, cancel units before calculating, round once at the end, and use repeated trials when discussing precision.

SUPPORT

Keep the prefix table, factor-label frame, and formula bank visible. Let students annotate units in color. Reduce arithmetic load without removing the reasoning step.

EXTEND

Ask students to design a repeated-trial measurement plan, predict the effect of a systematic offset, and defend which additional data would strengthen an accuracy claim.

ASSESSMENT USE

Use Form A after instruction and mixed review. Reserve Form B for retakes, an alternate class period, or test security. Do not use either test as the first exposure to a skill.

SCOPE

Supports the NGSS practice of Using Mathematics and Computational Thinking; no formal performance-expectation alignment is claimed.